

Project Design Phase
Solution Architecture

Date:	31 July 2026
Name:	Kesireddy Padma
Project Name:	Metro Ticket Generating System in ServiceNow
Maximum Marks:	4 Marks

Solution Architecture Description

Solution architecture is the foundational framework that connects the metro ticket booking requirements with the automated ServiceNow solution. For this system, the architecture maps a secure, validated, and transparent data flow from passenger ticket requests to automated ticket generation. Its core structural goals include:

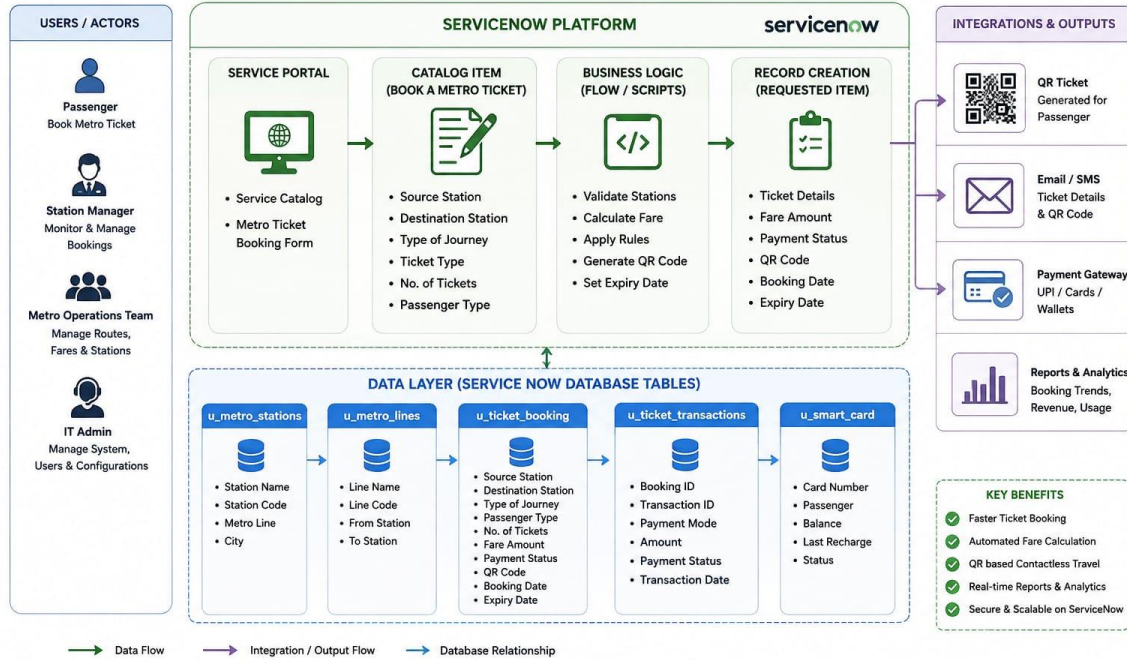
- Solving Existing Business Friction: Reducing long queues, manual ticket processing, fare calculation errors, and delays in metro ticket generation.
- Providing Stakeholder Clarity: Defining the relationships between passengers, metro stations, ticket details, payment status, and QR-based ticket generation.
- Defining System Requirements: Outlining the execution flow across the Service Catalog, client-side validation, Flow Designer automation, station data, and ticket records.
- Enforcing Technical Standards: Applying validation rules, access controls, data integrity checks, automated fare calculation, payment validation, and secure ticket generation.

Solution Architecture and Data Flow Diagram

The end-to-end architecture maps how passenger input from the ServiceNow ticket booking form is validated and transformed into a structured metro ticket record with fare, payment, QR code, booking date, and ticket expiry information.

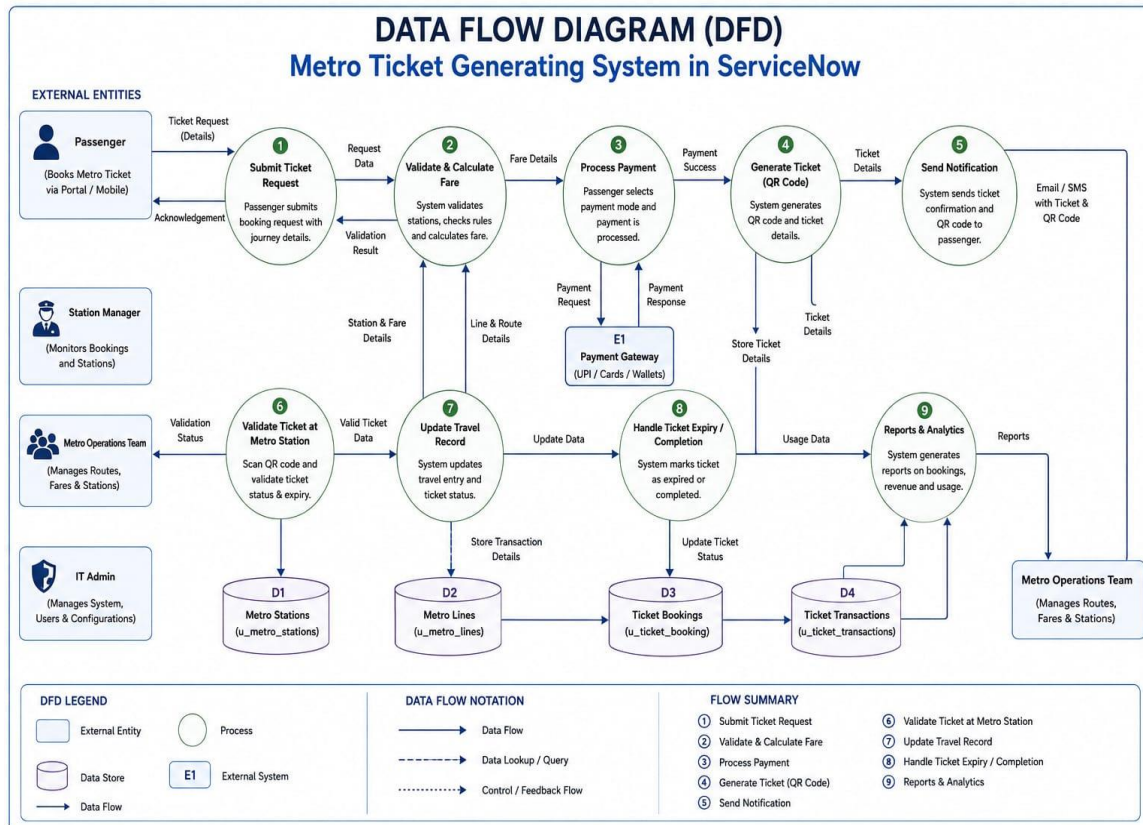
SOLUTION ARCHITECTURE

Metro Ticket Generating System in ServiceNow



DATA FLOW DIAGRAM (DFD)

Metro Ticket Generating System in ServiceNow



Architectural Data Flow Breakdown

The solution architecture uses a multi-stage pipeline organized into four logical runtime phases to process, validate, approve, and generate metro ticket transactions:

1. Passenger Ticket Request (Validated Front-End UI Layer)

The process starts when a passenger opens the ServiceNow Service Catalog item for booking a metro ticket. The passenger selects the source station, destination station, passenger type, number of tickets, and ticket type. Client-side logic validates the input and dynamically manages relevant fields.

2. Fare Calculation and Process Orchestration (ServiceNow Flow Designer)

After the request is submitted, ServiceNow automation processes the ticket request. Station details are mapped using the Metro Station Details table, and the fare is calculated based on the selected source, destination, passenger details, and number of tickets. Flow Designer coordinates the backend workflow and updates the requested item with the required ticket information.

3. Payment Validation and Ticket Generation

The system validates the payment status before completing the ticket request. Once payment is successful, the workflow proceeds with ticket generation. The system records the booking date, ticket expiry, fare, and payment status and generates a QR code that can be used for ticket verification.

4. Secure Ticket Record and Completion Receipt

After successful validation and processing, the final ticket details are stored in ServiceNow. The passenger receives a completion confirmation containing the generated ticket and QR code. The ticket can then be used for metro entry and verification within its validity period.